JointGuard: dependence diagnostics for probabilistic elastic-response predictions

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Abstract

Can correct displacement marginals conceal a wrong prediction of spring stress? An original two-degree-of-freedom elastic example isolates this question with Gaussian response forecasts. We compare marginal continuous ranked probability scores, joint logarithmic and energy scores, and a first-order variogram score. Population expectations are evaluated analytically or by one-dimensional quadrature, while a C++17 ensemble scorer is checked independently. An independence approximation preserves every displacement marginal but predicts an absolute stress exceedance probability 18.92 times the declared truth. Conversely, a common displacement bias leaves both the variogram score and central-spring stress risk unchanged. A finite-ensemble experiment separates sampling variability from numerical error. The study provides reproducible diagnostics for a synthetic model, not experimental validation or a new scoring rule.

1 Question and mechanical model

Probabilistic scientific models often produce joint predictions of spatial responses. Marginal diagnostics do not determine the dependence between those responses. This study asks whether that omission matters for a simple internal force, and whether a dependence-sensitive score alone is sufficient. Proper scoring rules supply a principled loss framework [1]; variogram scores provide a complementary view of dependence [2].

Consider two axial displacements $u = (u_1, u_2)^T$, each connected to ground by a spring and connected to each other by a third spring. All stiffnesses are $k = 10^6\,\mathrm{N/m}$. Quasi-static, small-displacement, linear equilibrium is

$$Ku = f, \qquad K = k\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}, \qquad f \sim \mathcal{N}(0, \Sigma_f), \quad \Sigma_f = K(s^2 C_{0.9})K^T, \quad C_\rho = \begin{bmatrix} 1 & \rho \\ \rho & 1 \end{bmatrix}, \tag{1}$$

where $s = 10^{-3}\,\mathrm{m}$. This deliberately prescribed load covariance produces the dimensionless truth $Y = u/s \sim \mathcal{N}(0, C_{0.9})$. The central-spring force is $k(u_2 - u_1)$. Assigning a nominal cross-sectional area $A = 10^{-4}\,\mathrm{m}^2$ defines a stress proxy $\sigma = ks(Y_2 - Y_1)/A$. The event $|\sigma| > 10\,\mathrm{MPa}$ is precisely $|Y_2 - Y_1| > 1$. This is a chosen reporting threshold, not a measured yield limit. For $D = a^T X$, $a = (-1, 1)^T$, under any Gaussian forecast,

$$P(|D| > b) = \Phi\left(\frac{-b - \mu_D}{s_D}\right) + \Phi\left(\frac{\mu_D - b}{s_D}\right), \qquad \mu_D = a^T\mu, \quad s_D^2 = a^T\Sigma a. \tag{2}$$

No temporal integration, mesh or fitted parameter is involved. All scores use the same fixed dimensionless displacement scale; their magnitudes cannot be compared as if they had identical units or normalization.

2 Scoring and numerical algorithm

Let F be the forecast law, $X, X' \sim F$ independent and y an observed response. Losses are oriented so that lower is better. The marginal score is the average of two univariate CRPS values. Joint scores are

$$\mathrm{LogS}(F, y) = -\log f_F(y), \tag{3}$$

$$\mathrm{ES}(F, y) = \mathbb{E}\|X - y\|_2 - \tfrac{1}{2}\mathbb{E}\|X - X'\|_2, \tag{4}$$

$$\mathrm{VS}(F, y) = (|y_1 - y_2| - \mathbb{E}|X_1 - X_2|)^2. \tag{5}$$

The library generalizes VS to the average over unordered coordinate pairs (order one, equal weights). Uniform shifts of the forecast cancel in each pairwise difference, so VS is not strictly proper for the full joint law. ES remains sensitive to dependence in population; this work does not label it blind merely because its excess loss is smaller than another score's.

For M independent ensemble members, the C++ implementation evaluates

$$\widehat{\mathrm{ES}}_U = \frac{1}{M}\sum_{i=1}^{M}\|x_i - y\|_2 - \frac{1}{M(M-1)}\sum_{i<j}\|x_i - x_j\|_2. \tag{6}$$

This estimates the population score without finite-ensemble bias when members are iid and independent of y. Replacing $M(M-1)$ by M^2 instead computes the score of the finite empirical distribution, with expected excess $\mathbb{E}\|X - X'\|_2/(2M)$ over the population score. Neither denominator is universally correct: it depends on which distribution is being evaluated. The iid correction is not justified for correlated ensemble members.

For independent Gaussian forecast and truth, the two norms in the population ES have Gaussian arguments. For $Z \sim \mathcal{N}(m, S)$, define

$$L(t) = \mathbb{E}e^{-tZ^TZ} = \frac{\exp[-tm^T(I+2tS)^{-1}m]}{\sqrt{\det(I+2tS)}}. \qquad \mathbb{E}\|Z\|_2 = \frac{1}{\sqrt{\pi}}\int_0^\infty \frac{1-L(x^2)}{x^2}\,dx. \tag{7}$$

The integral follows by applying the Laplace representation of $\sqrt{z}$ to Z^TZ and substituting $t = x^2$. We diagonalize S, normalize the scale, use `log1p`/`expm1` to avoid subtraction cancellation, and apply adaptive quadrature with absolute and relative tolerances 2×10^{-11} in normalized units. Its error estimate is not a rigorous certificate.

Closed-form folded-normal moments give $\mathbb{E}|Z|$ and therefore population CRPS and VS. Gaussian expected LogS follows from a trace term, a squared mean-error term and a log determinant. Cholesky solves avoid explicit covariance inverses. Force moments propagate as $\mu_u = K^{-1}\mu_f$ and $\Sigma_u = K^{-1}\Sigma_f K^{-T}$. Inputs must be finite with symmetric positive-definite covariance and stiffness; singular laws are not supported.

3 Implementation and reproducible experiment

The repository contains a C++17 distance core, pybind11 bindings, and a Python package for Gaussian diagnostics, elastic propagation and experiments. For N observations in d dimensions, the native scorer costs $O((M^2 + NM)d)$ operations and $O(N)$ auxiliary storage; it does not allocate a pairwise distance matrix. No measured runtime speedup is claimed.

Four forecasts are specified without training: calibrated $\mathcal{N}(0, C_{0.9})$, independent $\mathcal{N}(0, I)$, anticorrelated $\mathcal{N}(0, C_{-0.5})$, and common bias $\mathcal{N}((1,1)^T, C_{0.9})$. A 64-point correlation sweep on $[-0.9, 0.99]$ retains unit marginal variances. The stochastic check uses NumPy's default random generator with seed 20261020, 24 independent replicates, 128 ensemble members and 256 independent observations per replicate. Within each replicate the candidates share latent Gaussian draws for paired comparison; observations and forecast ensembles are independent. Standard errors are computed across replicate means and include both ensemble and observation sampling.

The command `jointguard -output results` regenerates the experiment.

Run `scripts/check_results.py` with Python for an independent repeat. JSON metrics, three CSV tables, the figure, and LaTeX tables/macros share this experiment. The paper imports these artifacts directly. The build script rejects unresolved references, overfull boxes and more than five pages, and records SHA-256 hashes of every paper input and the PDF. CI tests Windows and Ubuntu builds and independently compiles the paper with pdfLaTeX.

4 Results and validation

Table 1 shows that the three zero-mean forecasts have identical marginal CRPS, despite substantially different extension risk. Ignoring the positive response correlation raises the predicted exceedance probability by a factor of 18.92. Joint LogS, ES and VS each identify this dependence error in population. A uniform mean

Forecast	CRPS	LogS	ES	VS	Risk (%)
Calibrated	0.5642	2.0075	0.8267	0.0727	2.535
Independent	0.5642	2.8379	0.8626	0.6680	47.950
Anticorrelated	0.5642	3.6274	0.8965	1.1236	56.370
Common bias	0.8351	2.5338	1.2049	0.0727	2.535

Table 1: Expected loss under the declared truth, and forecast probability of absolute stress exceedance. ES uses deterministic quadrature; all other columns are closed form. Values are generated, not estimates from the finite sample.

shift instead leaves VS and central-spring risk unchanged while increasing marginal CRPS, LogS and ES. Being irrelevant to this particular stress event does not make that bias irrelevant to displacement predictions or the two grounding-spring forces.

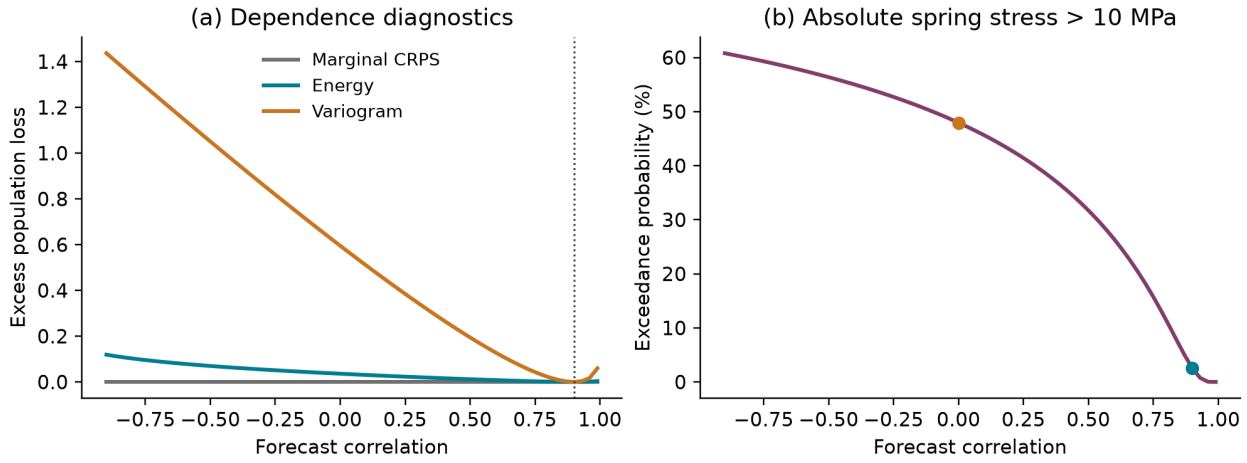


Figure 1: Original correlation sweep. (a) Excess loss relative to the calibrated forecast; the dotted line marks true correlation. Different losses have different scales, so their vertical separation is not a universal sensitivity ranking. (b) Predicted absolute stress exceedance; dots mark calibrated and independent forecasts. Marginals stay unchanged throughout the sweep.

For the calibrated forecast, the finite-ensemble mean unbiased ES is 0.82363 with replicate standard error 0.00670, compared with population ES in Table 1. The observed empirical-minus-unbiased difference is 0.00635, versus expected bias 0.00646 for $M = 128$. These are finite-sample results, not quadrature errors; the 24-replicate standard error is itself estimated. The reproduction check requires each model's ensemble mean to fall within four estimated standard errors of its population ES. This fixed-seed regression check does not establish a general confidence guarantee.

Independent numerical checks compare native scores against SciPy distance matrices (maximum absolute error `6.66e-16`), verify the equilibrium covariance identity (relative Frobenius residual `4.90e-16`), and compare the isotropic two-dimensional norm with its Rayleigh mean (absolute error `2.22e-16`). These are roundoff-scale observations, not claims of portable last-digit accuracy. Tests additionally cover noncentral one-dimensional norm quadrature, chi-distribution means, Gaussian log densities, invariance properties, input rejection and native overflow.

5 Discussion, limitations and conclusion

This example separates three questions. *Discrete implementation error* is checked by algebraic identities and independent distance calculations. *Numerical integration error* enters population ES through quadrature; the other population values are closed form. *Continuum error* is not estimated: the three-spring model is the stipulated discrete system, and no continuum discretization or mesh-convergence claim is made. Sampling uncertainty is reported separately for finite ensembles. None of these tests establishes *physical validity*; no measured material data, nonlinear constitutive response, inertia, failure model or experimental stress observations are used.

The findings apply to fixed, nonsingular Gaussian laws in a common coordinate scale. Non-Gaussian, high-dimensional, time-dependent and data-fitted forecasts could behave differently. The stress threshold illustrates a decision-relevant quantity but does not define a design safety assessment. Score comparisons alone do not supply tail calibration, and numerical loss magnitudes cannot be ranked across score families without considering their scales and targets.

The reproducible result is a complementary diagnostic workflow: assess marginals, evaluate the joint law, and inspect the actual mechanical functional of interest. Correct marginals can conceal large extension-risk errors, while a variogram can conceal a common displacement bias. Neither diagnostic should be used alone. The project was motivated by DTU ATSA lecture topics on multivariate forecast evaluation; no private course material is redistributed.

References

[1] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi:10.1198/016214506000001437.

[2] M. Scheuerer and T. M. Hamill. Variogram-based proper scoring rules for probabilistic forecasts of multivariate quantities. *Monthly Weather Review*, 143(4):1321–1334, 2015. doi:10.1175/MWR-D-14-00269.1.